

⑥ Verify that the following is a distribution function:

$$F(x) = \begin{cases} 0 & x < -a \\ \frac{x}{2a} + \frac{1}{2} & -a \leq x \leq a \\ 1 & x > a \end{cases}$$

Solution: ① $0 \leq F(x) \leq 1$, $-\infty < x < \infty$

True

② $F'(x) = f(x) \geq 0$ ($f(x)$ is P.d.f)

i.e $F(x)$ is non decreasing function of x

$$f(x) = F'(x) = \begin{cases} \frac{1}{2a} & -a \leq x \leq a \\ 0 & \text{otherwise} \end{cases}$$

clearly $f(x) \geq 0 \quad \forall x$

③ $F(-\infty) = 0$, true.

④ $F(\infty) = 1$, true

⑤ To verify $f(x)$ is P.d.f

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_{-a}^a \frac{1}{2a} dx \\ &= \frac{1}{2a} [x]_{-a}^a = \frac{1}{2a} (2a) = 1 \end{aligned}$$

Hence $F(x)$ is distribution function.

7) Let X be a Continuous R.V with P.d.f is

$$f(x) = \begin{cases} Kx & 0 \leq x < 1 \\ K & 1 \leq x < 2 \\ 3K - Kx & 2 \leq x < 3 \\ 0 & \text{else.} \end{cases}$$

a) Find K

b) Find $F(x)$

c)

c) if x_1, x_2 and x_3 are 3 independent observations from X , what is the probability that exactly one of these 3 numbers is larger than 1.5?

Solution

a) $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\int_0^1 Kx dx + \int_1^2 K dx + \int_2^3 (3K - Kx) dx = 1$$

$$K \int_0^1 x dx + K \int_1^2 dx + K \int_2^3 (3 - x) dx = 1$$

So we get $K = \frac{1}{2}$

b)

$$f(x) = \begin{cases} \frac{x}{2} & 0 \leq x < 1 \\ \frac{1}{2} & 1 \leq x < 2 \\ \frac{3}{2} - \frac{x}{2} & 2 \leq x < 3 \\ 0 & \text{else.} \end{cases}$$

By definition, $F(x) = \int_{-\infty}^x f(x) dx$

$\rightarrow \underline{x < 0}$: $F(x) = \int_{-\infty}^x f(x) dx = \int_{-\infty}^x 0 dx = 0$

for $x < 0$, $F(x) = 0$

$$\begin{aligned}
 \rightarrow \underline{0 \leq x < 1}: \quad F(x) &= \int_{-\infty}^x f(x) dx \\
 &= \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx \\
 &= 0 + \int_0^x \frac{x}{2} dx \\
 &= \left[\frac{x^2}{4} \right]_0^x = \frac{x^2}{4}
 \end{aligned}$$

$$\boxed{\text{for } 0 \leq x < 1, \quad F(x) = \frac{x^2}{4}}$$

$$\begin{aligned}
 \rightarrow \underline{1 \leq x < 2}: \quad F(x) &= \int_{-\infty}^x f(x) dx \\
 &= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^x f(x) dx \\
 &= 0 + \int_0^1 \frac{x}{2} dx + \int_1^x \frac{1}{2} dx \\
 &= \left(\frac{x^2}{4} \right)_0^1 + \frac{1}{2} (x)_1^x \\
 &= \frac{1}{4} + \frac{x-1}{2} = \frac{2x-1}{4}
 \end{aligned}$$

$$\boxed{\text{for } 1 \leq x < 2, \quad F(x) = \frac{2x-1}{4}}$$

$$\begin{aligned}
 \rightarrow \underline{2 \leq x < 3}: \quad F(x) &= \int_{-\infty}^x f(x) dx \\
 &= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^x f(x) dx \\
 &= 0 + \frac{1}{4} + \int_1^2 \frac{1}{2} dx + \int_2^x \left(\frac{3-x}{2}\right) dx \\
 &= \frac{1}{4} + \frac{1}{2}(2-1) + \frac{1}{2} \left[3x - \frac{x^2}{2} \right]_2^x \\
 &= \frac{1}{4} + \frac{1}{2} + \frac{1}{2} \left[3(x-2) - \frac{1}{2}(x^2-4) \right] \\
 &= -\frac{x^2}{4} + \frac{3x}{2} - \frac{5}{4}
 \end{aligned}$$

$$\text{for } 2 \leq x < 3, \quad F(x) = -\frac{x^2}{4} + \frac{3x}{2} - \frac{5}{4}$$

$$\begin{aligned}
 \rightarrow x \geq 3, \quad F(x) &= \int_{-\infty}^x f(x) dx \\
 &= 1
 \end{aligned}$$

$$\text{for } x \geq 3, \quad F(x) = 1$$

$$\therefore F(x) = \begin{cases} 0 & x < 0 \\ \frac{x^2}{4} & 0 \leq x < 1 \\ \frac{2x-1}{4} & 1 \leq x < 2 \\ -\frac{x^2}{4} + \frac{3x}{2} - \frac{5}{4} & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

(c) The Probability that X is larger than 1.5 is

$$\begin{aligned}P(X > 1.5) &= 1 - P(X \leq 1.5) \\&= 1 - F(1.5) \\&= 1 - \frac{2(1.5) - 1}{4} \\&= \frac{1}{2}\end{aligned}$$

The Probability that X is not larger than 1.5 is

$$\begin{aligned}P(X < 1.5) &= 1 - P(X > 1.5) \\&= 1 - \frac{1}{2} = \frac{1}{2}\end{aligned}$$

Hence, out of three numbers x_1, x_2 and x_3 , the Probability that exactly one is larger than 1.5 is

$$\begin{aligned}&= P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C) \\&= P(A)P(\bar{B})P(\bar{C}) + P(\bar{A})P(B)P(\bar{C}) + P(\bar{A})P(\bar{B})P(C) \\&= \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^3 \\&= \frac{3}{8}\end{aligned}$$

⑧ A Petrol pump is supplied with petrol once a day. If its daily volume of Sale X (in thousands of litres) is described by $f(x) = 5(1-x)^4$, $0 \leq x \leq 1$. What must be the Capacity of its tank in order that the Probability that its Supply will be exhausted in a given day is 0.01?

Solution Let the Capacity of the tank (in 1000's of litres) be a
Such that $P(X \geq a) = 0.01$

$$\int_a^1 f(x) dx = 0.01$$

$$\text{Solve, we get } a = 1 - \left(\frac{1}{100}\right)^{\frac{1}{5}} = 0.6019$$

Hence the Capacity of the tank is 601.9 litres.